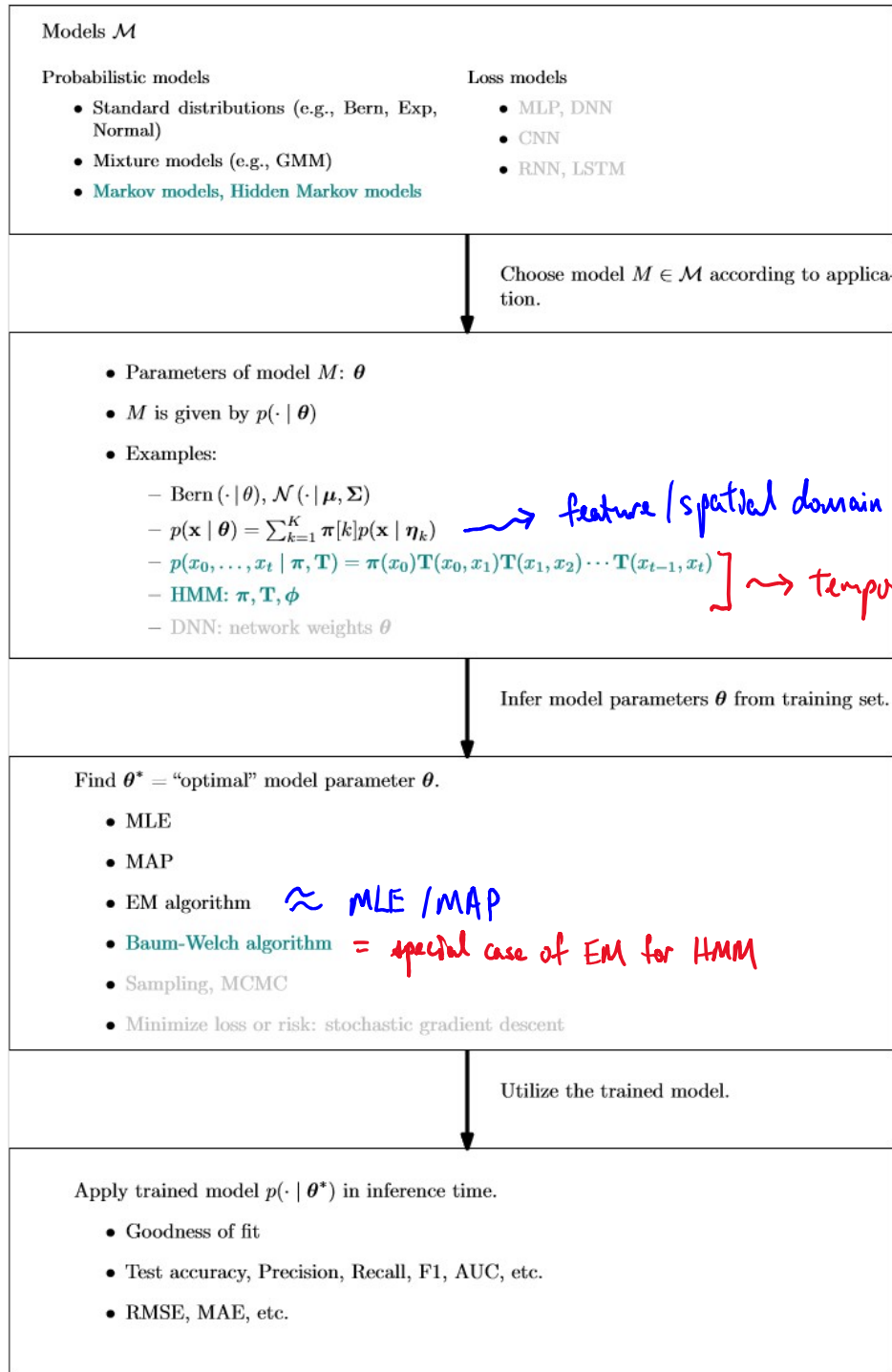




overview
week 4



Markov Chain

$x[0], x[1], x[2], \dots$

↖ "time" index
 $x_i[t]$
↖ sample index

$x[0], x[1], x[2], \dots$

$x[t] \in \text{discrete state space}$
 $\{1, 2, \dots, M\}$

τ sample index

$x_1 = (x_1[0], x_1[1], x_1[2], \dots, x_1[t])$

$x_2 = (x_2[0], x_2[1], \dots, x_2[t])$

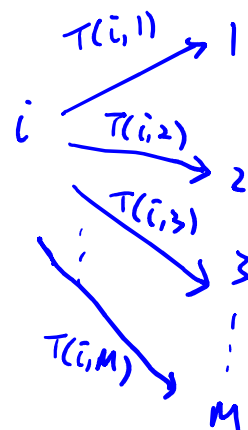
$$p(x[t] | x[0], x[1], \dots, x[t-1]) = p(x[t] | x[t-1]) \quad \text{Markov chain}$$

Transition probability $T(i, j) = p(x[t] = j | x[t-1] = i)$
 $i, j \in \{1, 2, \dots, M\}$

$x[0] \xrightarrow{T} x[1] \xrightarrow{T} x[2] \rightarrow \dots$

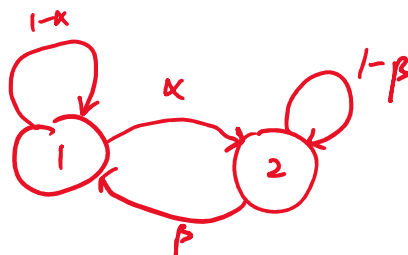
homogeneous MC

transition Matrix $T = \begin{bmatrix} T(1,1) & T(1,2) & \dots & T(1,M) \\ T(2,1) & T(2,2) & \dots & T(2,M) \\ \vdots & \vdots & \ddots & \vdots \\ T(M,1) & \dots & \dots & T(M,M) \end{bmatrix}$



$$\sum_{j=1}^M T(i, j) = 1 \quad \text{row-stochastic matrix}$$

$$T = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix} \end{matrix}$$



$x[0]$
 $\uparrow$
 $P_0 = [P_0(1), P_0(2), \dots, P_0(M)]$
 $P(x[0]=1)$
 $P(x[0]=2)$

$x[1]$

$P_i(i) = P(x[1] = i)?$
 $i = 1, \dots, M$

$P_i(i) = P(x[1] = i)$

$$p_i(i) = \mathbb{P}(x[1] = i)$$

$$= \sum_{\bar{j}=1}^M \mathbb{P}(x[1] = i, x[0] = \bar{j})$$

$$= \sum_{\bar{j}=1}^M \underbrace{\mathbb{P}(x[0] = \bar{j})}_{p_0(j)} \underbrace{\mathbb{P}(x[1] = i | x[0] = \bar{j})}_{T(j, i)}$$

$$= \sum_{\bar{j}=0}^M p_0(j) T(j, i)$$

$$= p_0 T(i)$$

$$\begin{bmatrix} \text{---} \\ p_0 \end{bmatrix} \begin{bmatrix} \text{---} \\ T \end{bmatrix} = \begin{bmatrix} \text{---} \\ \overset{(i)}{\downarrow} \end{bmatrix}$$

$$\underline{p_1 = p_0 T}$$

$$x[2] ? \quad p_2 = p_1 T = p_0 T T = p_0 T^2$$

$$p_3 = p_0 T^3$$

$\vdots$

$$p_t = p_0 T^t$$

state space = $\{1, 2\}$

$$(x[0], x[1], x[2]) = (1, 2, 1)$$

$$\underbrace{p(x[0]=1, x[1]=2, x[2]=1)} = \underbrace{p(x[0]=1)} \underbrace{p(x[1]=2 | x[0]=1)}$$

$$\cdot \underbrace{p(x[2]=1 | \cancel{x[0]=1}, \cancel{x[1]=2})}$$

$$= p_0(1) T(1, 2) T(2, 1)$$

$$= 0.5 \times 0.8 \times 0.7$$

$$X = (X[0], X[1], \dots, X[t])$$

$$P(X | \pi, T) = \pi(X[0]) \overset{\text{transition matrix}}{T}(X[0], X[1]) T(X[1], X[2]) \dots T(X[t-1], X[t])$$

$\uparrow$
 initial distribution (p_0)
 $\pi(i) = P(X[0]=i)$

Training dataset

$$\mathcal{D} = \{ \underbrace{x_1[0:t_1]}_{\text{"}}, \underbrace{x_2[0:t_2]}_{\text{"}}, \dots, x_n[0:t_n] \}$$

$$(x_1[0], \dots, x_1[t_1]) \quad (x_2[0], \dots, x_2[t_2])$$

need not be same $t_1 \neq t_2$

$$i \in \{1, \dots, M\}$$

$$x_1[0] \quad x_1[1] \quad \dots \quad x_1[t_1]$$

$$x_2[0] \quad x_2[1] \quad \dots \quad x_2[t_2]$$

$$\vdots$$

$$x_n[0] \quad \dots \quad x_n[t_n]$$

log likelihood:

$$\log p(\mathcal{D} | \pi, T) = \sum_{i=1}^n \log p(x_i[0:t_i] | \pi, T)$$

$$= \sum_{i=1}^n \log \pi(x_i[0]) + \sum_{i=1}^n \sum_{t=1}^{t_i} \log T(x_i[t-1], x_i[t])$$

$$= \sum_{x=1}^M N_x \log \pi(x) + \sum_{x,y=1}^M N_{xy} \log T(x, y)$$

$$N_x = \# \text{ of seq. with } x_i[0] = x$$

$$= \sum_{i=1}^n \mathbb{1}_{\{x_i[0]=x\}}$$

$$N_{xy} = \# \text{ of transitions from } x \text{ to } y$$

$$= \sum_{i=1}^n \sum_{t=1}^{t_i} \mathbb{1}_{\{x_i[t-1]=x, x_i[t]=y\}}$$

MLE for $\pi(x)$: $\pi(x) \geq 0, \sum_x \pi(x) = 1$

$$1 - 0.1 - 0.1 - \dots - 0.5 - \dots$$

$$0.1 - 0.1 - \dots$$

MLE for $\pi(x)$: $\pi(x) > 0, \sum_x \pi(x) = 1$

$$\mathcal{L} = N_x \log \pi(x) - \lambda \sum_y \pi(y)$$

Supplementary.

$$\frac{\partial}{\partial \pi(x)} \mathcal{L} = \frac{N_x}{\pi(x)} - \lambda = 0$$

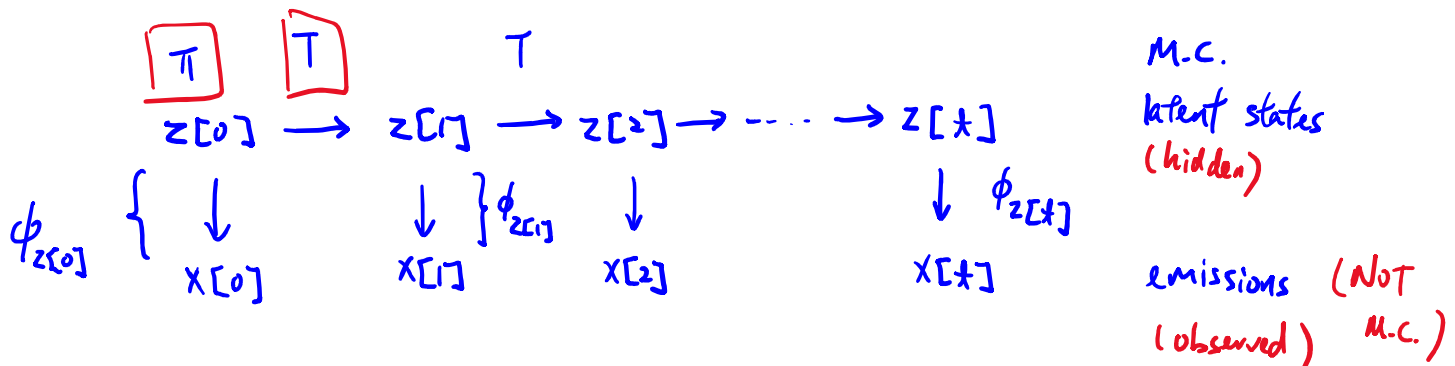
$$\pi(x) = \frac{N_x}{\lambda} = \frac{N_x}{n}$$

$$\sum_{x=1}^M \pi(x) = 1$$

$$\frac{1}{\lambda} \left[\sum_{x=1}^M N_x \right] = 1$$

$$\lambda = \frac{1}{n}$$

Hidden Markov Model (HMM)



$$p(x[t] | z[t]) = p(x[t] | \underbrace{\phi_{z[t]}}_{\phi_1, \phi_2, \dots, \phi_M}) \quad z[t] \in \{1, 2, \dots, M\}$$

E.g., $x[t] | z[t]=m \sim N(\mu_m, \Sigma_m)$

$$\begin{pmatrix} \mu_1, \Sigma_1 \\ \mu_2, \Sigma_2 \end{pmatrix}$$

... μ_m, Σ_m

$$\begin{pmatrix} (\mu_2, \Sigma_2) \\ \vdots \\ (\mu_M, \Sigma_M) \end{pmatrix}$$

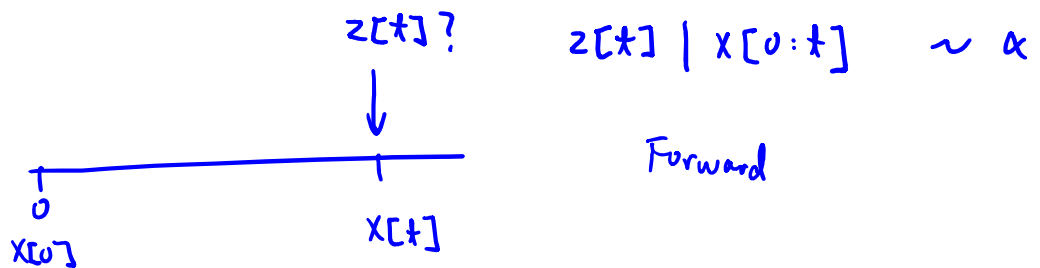
Training data

$$\mathcal{D} = \{x_1[0:t_1], \dots, x_n[0:t_n]\}$$

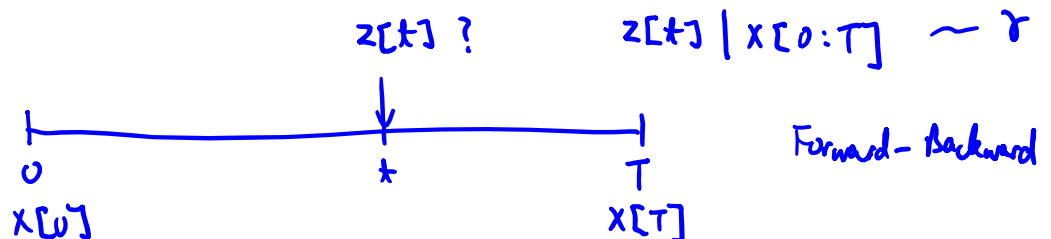
$$\text{easy } \tilde{\mathcal{D}} = \{(x_1[0:t_1], z_1[0:t_1]), \dots, (x_n[0:t_n], z_n[0:t_n])\}$$

$$\begin{aligned} \log p(\tilde{\mathcal{D}} | \pi, T, \phi) &= \sum_{i=1}^n \log \pi(z_i[0]) + \sum_{i=1}^n \sum_{t=1}^{t_i} \log T(z_i[t-1], z_i[t]) \\ &+ \sum_{i=1}^n \sum_{t=1}^{t_i} \log p(x_i[t] | \phi_{z_i[t]}) \end{aligned}$$

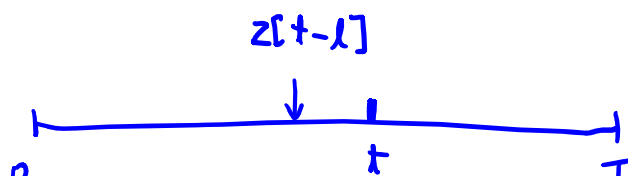
Fitting:

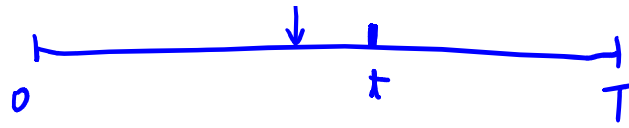


Smoothing:

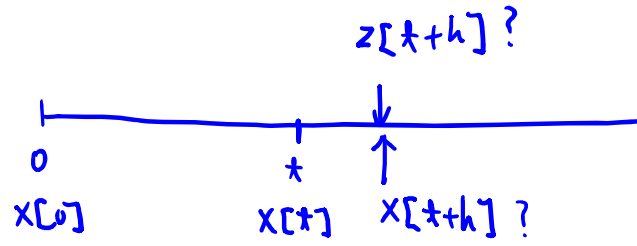


Fixed lag

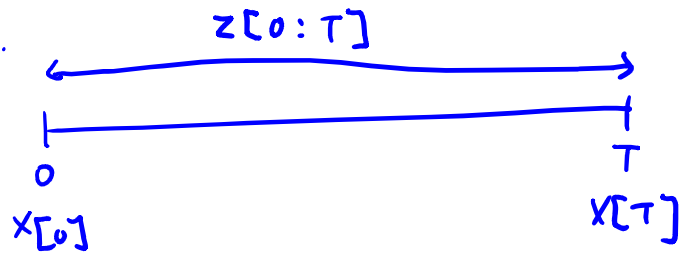




Prediction



MAP sequence.



Viterbi algorithm.